

Preliminary Analysis Concerning Morphometric data and Application for fishery Management on Cavala Preta *Decapterus macarellus*, in Cabo Verde Waters

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Abstract

Total of 895 Cavala preta, *Decapterus macarellus*, from nine islands were obtained between May and July 2002, using R/V Sinagoga. Base on eight morphometric measurements, differences were suggested, using the Multi-Dimensional Scaling (MDS) Method. A clear geographical pattern was observed from the northern to the southern islands except the position of the Maio and Boavista islands. The surplus production model was applied for northern and southern waters. MSY and effort at MSY from northern waters were 2213 t and 2893 days of operations, and those from southern waters were 593 t and 2130 days, respectively. Some management measures will be needed in northern waters for Cavala preta fishery.

Introduction

It is necessary to identify population for the fishery management. Cavala preta *Decapterus macarellus*, is the most important fishery resource in Cabo Verde. Previous studies on Cavala preta were done by several researchers (Carvalho, 1993; Jardin, 1995; and Almada, 1997). All studies considered as one unit of stock in Cabo Verde. The fishery management is preferable to each stock. There are various methods to identify fish stocks such as morphometric and meristic counts methods, isozim by the electrophoresis, tagging method, DNA analysis, and so on.

The objective of this study is to examine whether Cavala preta, *Decapterus macarellus*, in Cabo Verdian waters consist of one or more than one unit of stocks, using morphometric method. In a case, stocks are discriminated, statistical analysis will be carried out to see resource conditions for each stock in Cabo Verde.

Materials and Methods

Samples of Cavala preta were corrected form all island from Cabo Verde (Figure 1). Total of 895 Cavala preta, *Decapterus macarellus*, from nine islands in Cabo Verde were obtained between May and July 2002 (Table 1), using Sinagoga. Samples were obtained by the purse seine.

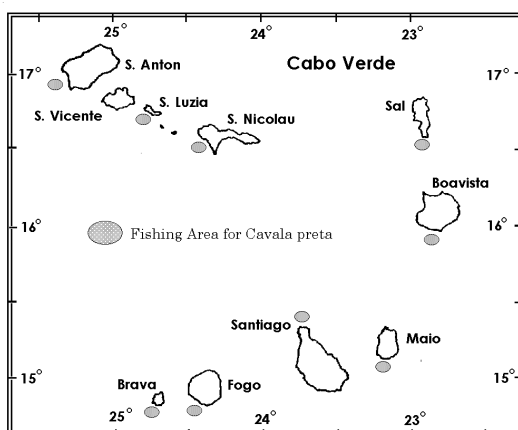


Figure 1. Sampling area for Cavala preta in 2002

Table 1. Summary for Cavala preta sampling in Cabo Verde.

Island	Abri. Island	Sample Collection	Place for Measurement	No of Samples			Range of FL (average),cm
				Female	Male	Total	
S. Antao	SA	July 5, 2002	INDP	40	60	100	24.9 - 32.2 (27.3)
S. Vicente	SV	May 25, 2002	INDP	59	41	100	24.3 - 29.9 (25.7)
S. Nicolau	SN	June 28, 2002	INDP	43	49	92	24.6 - 29.6 (27.1)
Sal	SAL	June 7, 2002	R/V Sinagoga	75	25	100	24.1 - 29.2 (26.4)
Boavista	BV	June 8, 2002	R/V Sinagoga	72	28	100	25.3 - 32.7 (28.2)
Maio	Maio	June 12, 2002	Praia Fishing Port	65	33	98	24.4 - 30.3 (26.2)
Santiago	ST	June 18, 2002	Praia Fishing Port	29	71	100	26.4 - 38.8 (31.8)
Fogo	Fogo	June 14, 2002	Port of S. Filipe	57	48	105	26.4 - 32.0 (28.6)
Brava	Brava	June 15, 2002	Port of Faja de Agua	57	43	100	26.1 - 31.5 (28.3)
Total				497	398	895	24.1 - 38.8 (27.8)

In principal, sample measurement was done wither on board of Sinagoga or nearest port. If the fishing ground was near S. Vicente Island, samples were taken to the INDP Laboratory. The morphometric measurements are shown in Figure 2. Eight characters were measured in the following:

1. FL: Fork Length
2. S1DF: Snout to insertion of first dorsal fin
3. S2DF: Snout to insertion of second dorsal fin
4. SAF: Snout to insertion of anal fin
5. SPA: Snout to insertion of pelvic fin
6. BH: Body height from the insertion of the first dorsal fin
7. HL: Head length
8. SPCF: Snout to insertion of the pectoral fin

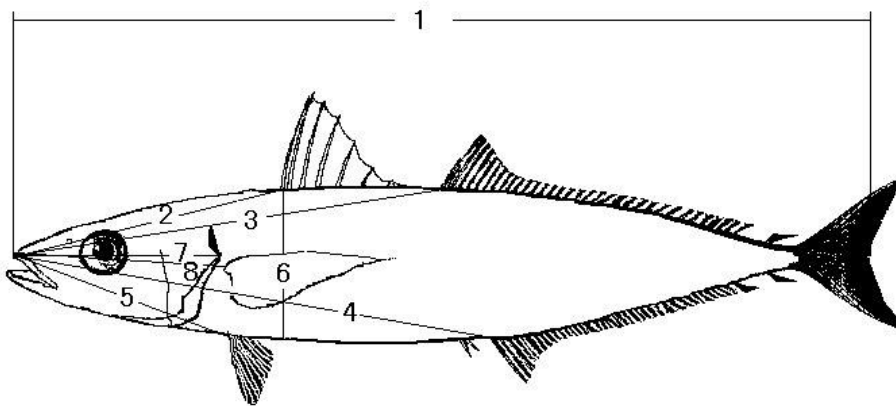


Figure 2. Morphometric characters to be measured for *Decapterus macurellus*.

The approach used was multivariate analysis concerning the 8 measurements undertaken for each individual. The statistical ordination method used was Multi-Dimensional Scaling (MDS) in its non-parametric form. Similarity between samples (individuals) was calculated using the Bray-Curtis similarity coefficient without any prior transformation of the data. In a case, MDS indicates more than two groupings, that is stocks, analysis of statistics will be carried out. Catch and effort data from the annual statistics of Cabo Verde will be utilized in order to see each stock, using the surplus production model (Schaefer, 1954).

Results

The resulting MDS plots are shown in Figure 3. From the results of the MDS, S. there are two groups: one is from S. Anton, S. Vicente, S. Nicolau, Sal, and Maio; the other Boavista,

Santiago, Fogo, and Brava. Maio and Boavista were geographically inverted, but overall results indicates two groups of *Cavala preta* in Northern (hereby Barlavento) and Southern (hereby Satavento) waters.

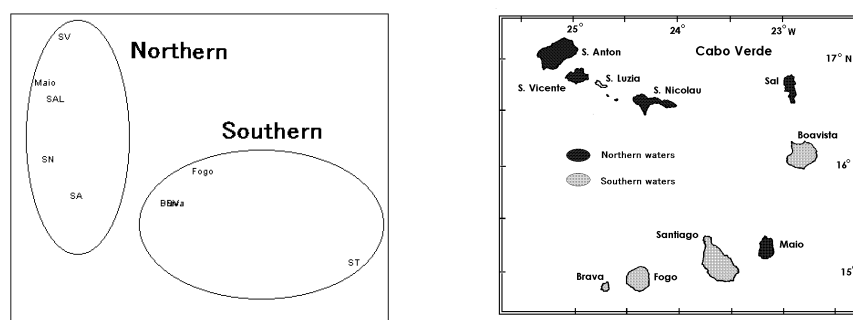


Figure 3. MDS plot based on average measurements by island.

As showing the result, two possible stocks in Barlavento and Satavento waters. In convenience of use of statistics, Barlavento is defined from S. Anton to Boavista, and Satavento is from Maio to Brava Islands.

Based on Annual Statistics in Cabo Verde, landings of *Cavala preta* from Artisanal and Industrial Fisheries were combined for each waters.

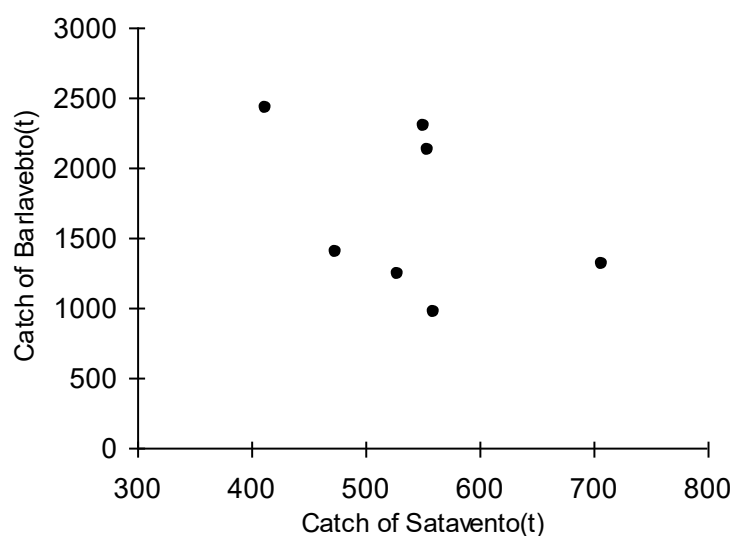


Figure 4. Relationship between Catch from Barlavento and Satavento from 1992 to 2000.

Figure 4 showed relationship between catch between Catch from Barlavento and atavento from 1992 to 2000. Although some negative relationship was identified, correlation coefficient of this linear relationship was rather weak ($r=0.41$). Landings from two waters of *Cavala preta*, thus, can be independent. The surplus production model was applied for these waters. Results were shown Figure 5, and Table 2.

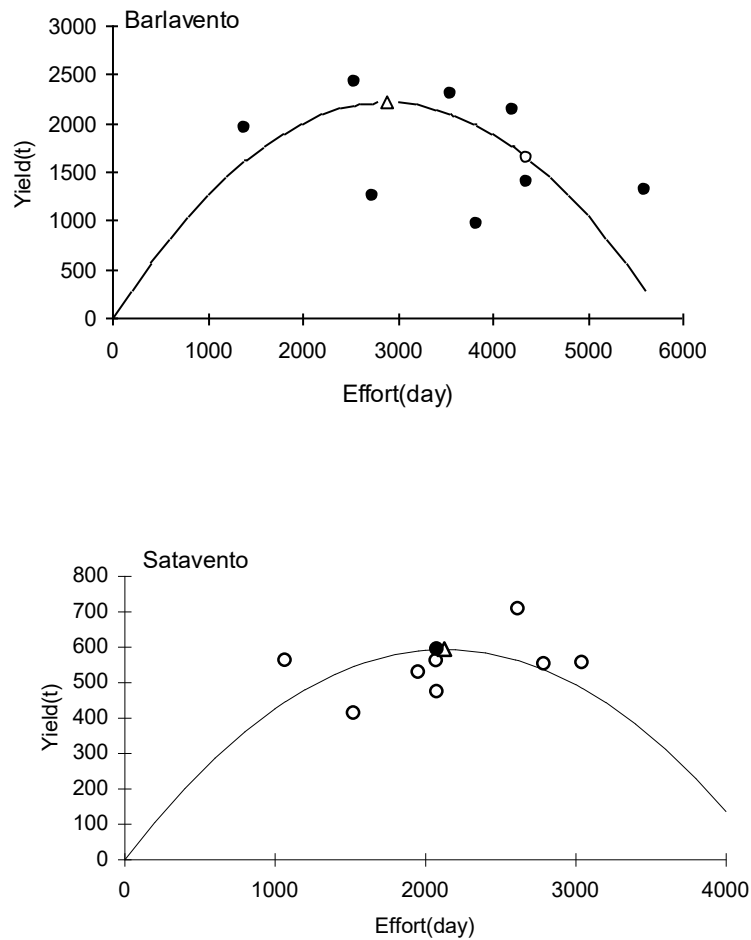


Figure 5. Surplus production curve for Barlavento and satabento, using data from 1992 to 2000.

Table 2. Estimated MSY and effort at MSY.

Area	MSY(t)	Effort at MSY(Day)
Barlavento	2213	2893
Satabento	593	2130

Discussions

In order to simplify, the MDS analysis was undertaken based on the average of the 8 eight morphometric measurements calculated for each island. As seen in Figure 3, a clear pattern was observed from the northern to the southern islands, which further reinforces the previous results based on individuals. Although the sample from Santiago appears to be extreme, and the position of the Maio and Boavista samples do not fit nicely in the north-south trend, overall trends can be some insight of differentiating stocks.

The pattern observed, which is a position in the MDS plot following a north-south gradient, does not showed a more distinct geographic grouping. This gradient may be more related to oceanographic conditions, resulting in differences in temperature and productivity. Furthermore, the schooling behaviour of Cavala preta may require analysis based on school samples.

The Surplus production application suggested that fisheries in Barlavento was over exploitation, while fisheries in Satavento was moderately exploited around MSY level. From the conservative management point of view, some counter-measurements of Cavala preta fishery in Barlavento should be made.

Literatures

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